Seismologist

Fervo Energy · Houston, TX

Job ID (this archive)	ENERGY-111
Employer	Fervo Energy
Location(s)	Houston, TX
Career stage (archive tag)	Graduate degree / research (Doctoral / Entry)
Location group	Texas (outside DFW)
Majors (archive tags)	Data Science, Computational & Geospatial Science
Energy area	Geothermal / geospatial and subsurface modeling
Pay listed	Not listed
Recruiting status when checked	Open when checked: the employer page had a live application. Checked Oct 26, 2026.
Source page	https://job-boards.greenhouse.io/fervoenergy/jobs/4431030009
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-082

Saved on 2026-10-26 from the employer's own careers page. The posting text below is the employer's own wording, reformatted for printing. Live listings change and close; check the source page before applying.

Job description

About Fervo Energy

Fervo Energy (NASDAQ: FRVO) is a modern power company developing next-generation geothermal resources to meet the growing demand for affordable, reliable, 24/7 carbon-free energy. Our mission is to transform geothermal into America's most dependable and affordable source of clean, around-the-clock power. By combining innovations in horizontal drilling, fiber-optic sensing, and advanced reservoir engineering, Fervo has developed a repeatable approach to building utility-scale enhanced geothermal systems. Our flagship Cape Station development in Utah is planned to deliver 500 MW of firm, carbon-free power, and we are advancing a development pipeline of more than 50 GW. Join us and work alongside world-class engineers, geoscientists, operators, project builders, and business leaders working to make geothermal a cornerstone of a more reliable, affordable, and sustainable energy future.

Position Description

Fervo is seeking a Seismologist to lead the design, deployment, and processing of surface and borehole seismic monitoring networks across our EGS development portfolio. You will own our arrays of three-component geophones and broadband seismometers — sited at the surface, in shallow vaults and boreholes, and downhole in monitoring wells — from network geometry and instrumentation through calibration, data quality, and continuous processing.

The central deliverable of this role is a high-precision, defensible earthquake catalog. Proficiency with HypoDD and GrowClust is required, and you will build, relocate, quality-control, and defend the catalogs that feed fracture-network characterization, reservoir models, induced-seismicity mitigation protocols, and regulatory and community engagement.

Fervo also operates fiber-optic Distributed Acoustic Sensing (DAS) in a number of wells, and you will incorporate DAS alongside the conventional sensor network as a complementary observing system. Familiarity with DAS is valuable, but it is not the focus of the role and is not required at hire — the conventional network and the relocated catalog come first.

Responsibilities

Surface and Borehole Geophone & Seismometer Networks

- Design, deploy, and maintain surface and borehole geophone and seismometer networks.
- Set network geometry and instrumentation to meet detection, location-accuracy, and completeness targets.
- Own instrument response, sensor orientation and calibration, timing, and continuous data QC.

- Run automated detection, phase picking, and event association across the network.
- Build and maintain 1D and 3D velocity models and station corrections.
- Manage field campaigns, vendor scopes, and data-delivery standards.

Earthquake Location, Relocation, and Catalog Development

- Run HypoDD for double-difference relocation.
- Run GrowClust for relative relocation and hierarchical clustering.
- Generate waveform cross-correlation differential travel times and maintain automated relocation pipelines.
- Determine source parameters — magnitude, focal mechanism, moment tensor, and stress drop.
- Interpret relocated seismicity as fracture geometry, stimulated volume, and fault reactivation.
- Maintain a versioned production catalog with documented magnitudes, completeness, and uncertainty.

Distributed Acoustic Sensing (DAS) — complementary observing system

- Incorporate DAS data from instrumented wells alongside the conventional network.
- Perform joint DAS and sensor processing where it improves the catalog.
- Interpret DAS response in a borehole setting, including its resolution limits.

Induced Seismicity, Hazard, and Risk

- Track b-value and magnitude of completeness, and forecast seismicity rates and maximum magnitude.
- Develop real-time hazard algorithms using physics-based, statistical, and machine-learning methods.
- Maintain traffic light protocols and mitigation strategies with operations.
- Link observed seismicity to Thermo-Hydro-Mechanical-Chemical reservoir models.
- Represent Fervo with regulators, partners, and communities on induced-seismicity questions.

Qualifications

Required

- Ph.D. in Geophysics, Seismology, Applied Mathematics, or a closely related field. No minimum years of industry or post-doctoral experience are required — recent and upcoming graduates are encouraged to apply.
- Working proficiency with HypoDD and GrowClust, including generation of waveform cross-correlation differential travel times, parameter selection, and assessment of relocation uncertainty. This may have been developed through thesis work, post-doctoral research, or industry practice.
- Solid grounding in surface and borehole seismic monitoring — network geometry and detectability, instrument response, sensor orientation and calibration, and earthquake detection, picking, and location.
- Knowledge of earthquake source analysis, including moment tensor inversion, focal mechanism determination, magnitude estimation, and stress drop estimation.
- Understanding of induced seismicity and seismic hazard concepts — magnitude of completeness, b-value, Gutenberg-Richter statistics, and ground-motion estimation.
- Python proficiency for seismic data processing (ObsPy, NumPy, SciPy, pandas, or equivalent tooling) and comfort working in Linux environments with shell scripting and version control.
- Ability to build velocity models and to produce, quality-control, and defend earthquake catalogs that others rely on.
- Willingness to travel to field sites for network deployment, commissioning, and troubleshooting up to 20%.

Preferred

- High Performance Computing and large-scale distributed processing of continuous seismic data.
- Machine learning applied to seismological problems, including deep-learning phase pickers.

Location

Fervo Energy has offices in Houston, TX, Golden, CO, Reno, NV, Oakland, CA, and Salt Lake City, UT. This position will be eligible for some hybrid work flexibility, but regular in-office presence at any of the above locations will be required.

Benefits & Perks

- Fervo offers a comprehensive benefits package designed to support your health, financial well-being, and life outside of work. Benefits for eligible employees include:
- Medical, dental, and vision plan options
- Employer-paid basic life insurance and short-term and long-term disability coverage
- A 401(k) plan with up to 6% match

- Flexible paid time off and paid parental leave
- Family-building, fertility, hormonal health, pregnancy, postpartum, and parenting support through Carrot
- A dependent care FSA for eligible child and elder care expenses
- An annual Lifestyle Spending Account for eligible wellness expenses, including fitness, financial planning, pet insurance, and pet care
- Employee ownership opportunities through Fervo's Employee Stock Purchase Program, as well as LTI and bonus opportunities for eligible roles

Benefits and programs are subject to eligibility requirements and may change.

Our Workplace

Fervo Energy highly values having employees work in the office to foster collaboration and strengthen our company culture. This role follows a hybrid schedule, with team members expected to work in the office three days per week. Fervo considers this in-office schedule an essential function of the role. Your recruiter can share more information about the schedule.

Equal Opportunity Employer

Fervo Energy is an Equal Opportunity Employer and does not discriminate on the basis of race, color, creed, gender, religion, marital status, registered domestic partner status, age, national origin, ancestry, physical or mental disability, medical condition, sex, genetic information, sexual orientation, military or veteran status, or any other consideration made unlawful by federal, state, or local law. Fervo also prohibits discrimination based on the perception that an individual has any of these characteristics or is associated with a person who has or is perceived as having any of these characteristics.

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: Ph.D. in Geophysics, Seismology, Applied Mathematics, or a closely related field. No minimum years of industry or post-doctoral experience are required — recent and upcoming graduates are encouraged to apply. | Ph.D. in Geophysics, Seismology, Applied Mathematics, or a closely related field. No minimum years of industry or post-doctoral experience are required — recent and upcoming graduates are encouraged to apply.

Key skills listed in the archive: Python; Linux; machine learning; statistics; geophysics; seismology

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Active · PDF: Complete · Source: Official employer Greenhouse Employer / ATS identifier: 4431030009

Career stage: Doctoral / Entry

Major fit: Direct geospatial/computational pathway; exact education and experience requirements retained. Energy-sector connection: GIS, seismic analysis or subsurface simulation for geothermal electricity generation.

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer Greenhouse. Source identifier: 4431030009.

Sources and retrieval details

Employer posting

https://job-boards.greenhouse.io/fervoenergy/jobs/4431030009

Retrieved source

https://boards-api.greenhouse.io/v1/boards/fervoenergy/jobs?content=true

Retrieved: 2026-10-26T16:33:09.471773+00:00 (UTC). Major fit, career stage and energy-sector connection are archive annotations; the employer text above is preserved from the cited source.